

Why Does AI Hallucinate?

Why language models state false things confidently, how bad it actually is, and what changed in 2026.

Updated 31 August 2026. Originally published 3 March 2026.

What changed in this update

The March version was built on 2025 benchmarks and 2025 models — GPT-4o, o3, Claude 3.7, Gemini 2.0 Flash — and it ended on an optimistic note: the best hallucination rate had fallen from 21.8% in 2021 to 0.7% in 2025, a 96% reduction, and the line was still going down.

That conclusion does not survive the 2026 data. Two things happened. The main benchmark was rebuilt to be harder, and the rates everyone had been quoting turned out to be a property of the easy test rather than of the models. And the 2026 flagship models — Claude Fable 5, Claude Opus 5, GPT-5.6 Sol, Grok 4.5 — got measurably more accurate *and* measurably worse at knowing when to keep quiet.

Everything in this update — every model, every number, every chart — is new. The sections on *why* it happens are substantially unchanged, because that research has held up.

What is an AI hallucination?

A hallucination is when a model produces something **false, misleading, or fabricated, and presents it as fact**. The word is borrowed from psychology, but *confabulation* describes it better: filling a gap with plausible-sounding content that has no factual basis behind it.

The canonical example is still *Mata v. Avianca* (2023), where a New York lawyer was sanctioned for filing a brief full of case citations ChatGPT had invented. What has changed is the frequency. Databases tracking AI-fabricated citations in court filings counted **10 cases in 2023**. By **August 2026 the count was 1,963**. The failure mode did not get rarer as the models got better; it got more common as more people used them.

Two benchmarks, two different answers

Any single number for "the hallucination rate" is close to meaningless, and this is the most important thing to understand about the 2026 figures. Published rates for the same model in the same month range from 3% to 92%. They are not contradicting each other. They are measuring different things.

Grounded summarization

The model is handed a document and asked to summarize it. Everything it needs is in front of it, so a hallucination is a clear failure. This is what **Vectara's HHEM leaderboard** measures.

Open-ended factual recall

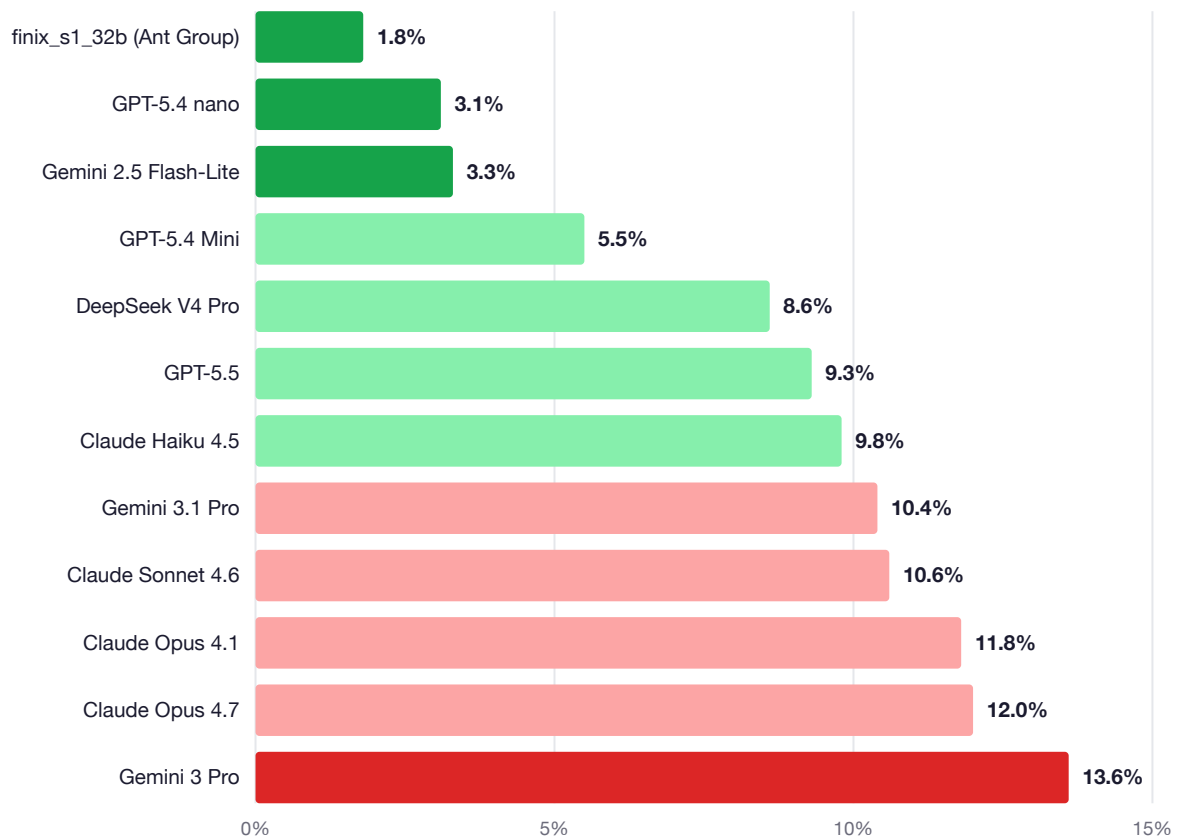
The model is asked a hard question with no source provided, and has to either know the answer or decline. This is what **AA-Omniscience** measures — and its rate is the share of questions the model

guessed on rather than admitting ignorance.

The same model can score 10% on the first and 60% on the second. Both numbers are correct.

Grounded summarization: Vectara HHEM

Vectara rebuilt this benchmark in November 2025: 7,700+ articles instead of 1,000, documents up to 32K tokens instead of short ones, spread across law, medicine, finance, technology and education. Rates on the new set are several times higher than on the old one — the models did not get worse, the test got closer to real work.

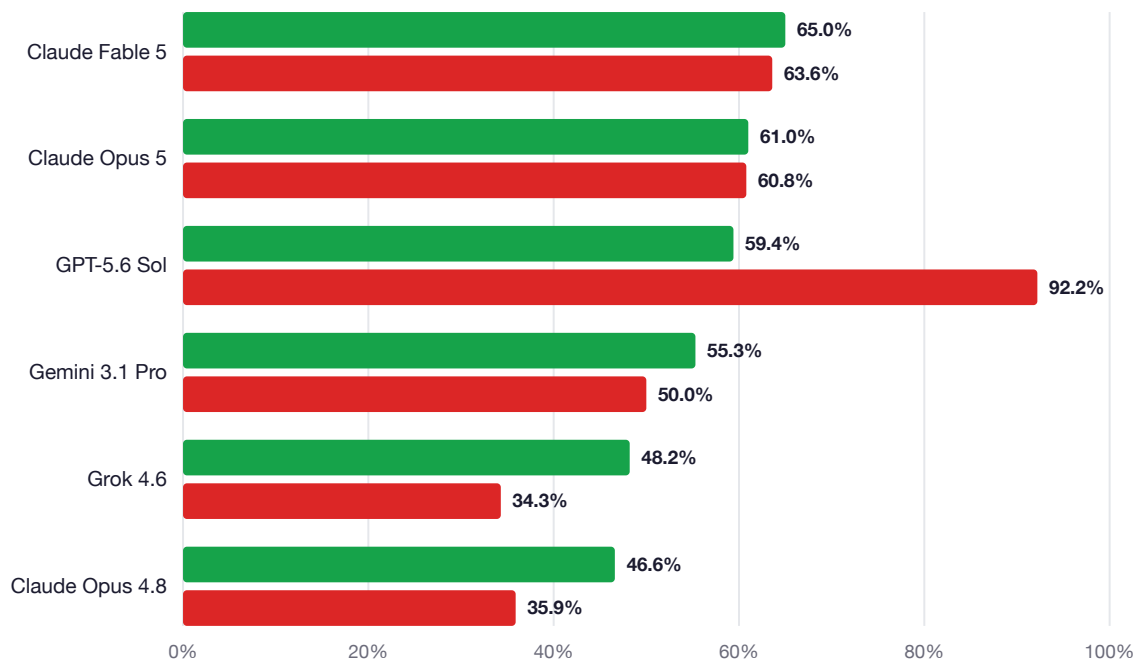


Vectara HHEM leaderboard, new dataset, as of 11 May 2026. Claude Opus 4.8, Claude Opus 5 and Claude Fable 5 have no HHEM score yet.

Note what is *not* at the top: the frontier models. The best scores belong to small and mid-sized models — a 32B model from Ant Group, GPT-5.4 nano, Gemini 2.5 Flash-Lite. Faithfully summarizing a document you have been given does not require a frontier model, and the frontier models are not the best at it.

Open-ended recall: the 2026 trade

Here is the finding that changed the story. The 2026 flagships are the most knowledgeable models ever measured. They are also, with one exception, the worst calibrated.

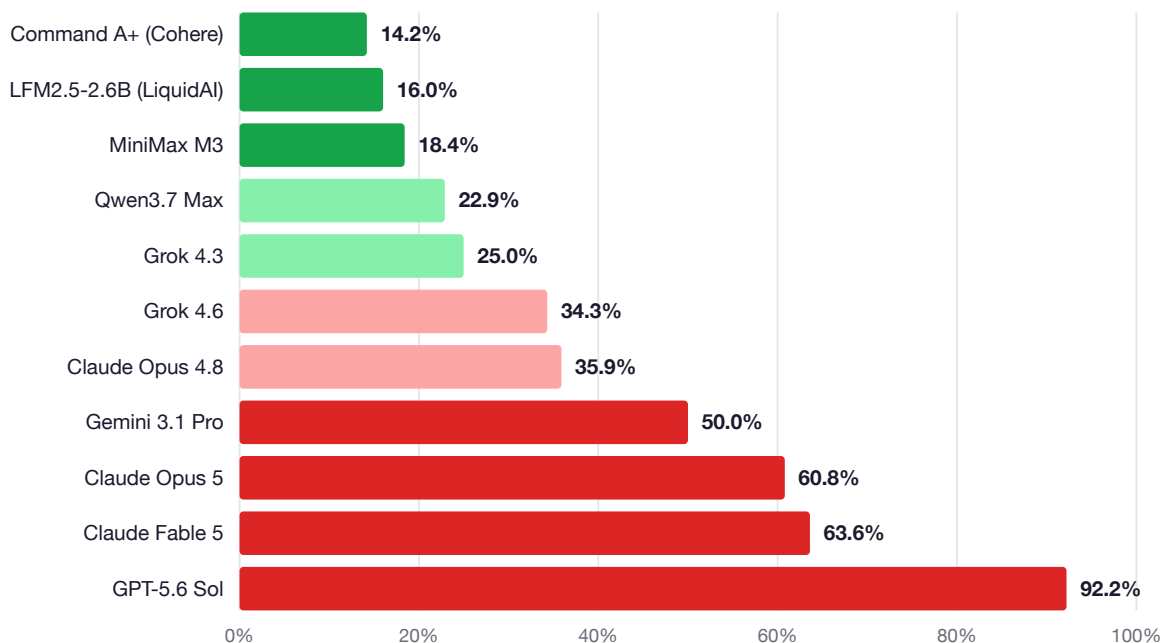


■ Accuracy — questions answered correctly ■ Hallucination rate — guessed instead of declining

AA-Omniscience, snapshot of 26–27 August 2026.

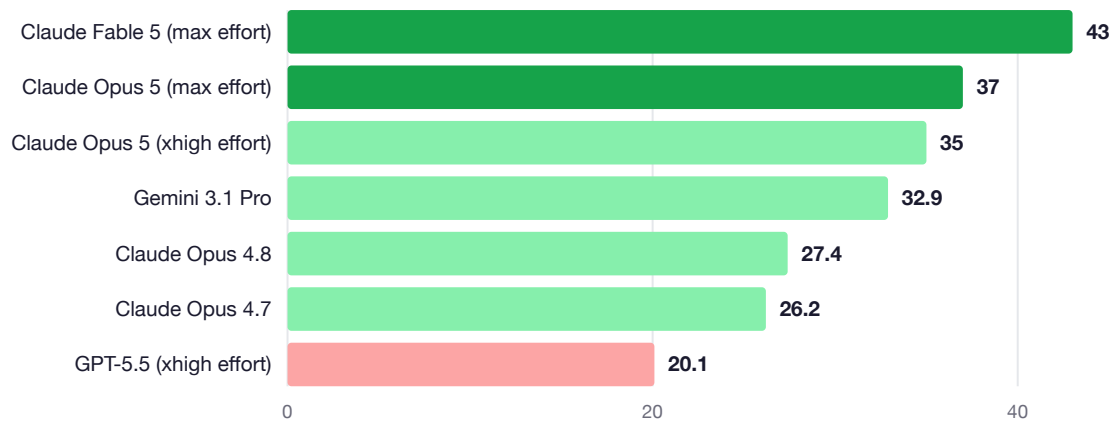
Read the top three rows again. Claude Fable 5 knows the most and guesses on 63.6% of what it does not know. GPT-5.6 Sol is close behind on accuracy and guesses on **92.2%** — it almost never says "I don't know". Grok 4.6 is the one model in the set that cut its hallucination rate without giving up accuracy.

Across the whole field the spread is enormous, and the models with the lowest rates are mostly not the ones anybody is using as a flagship:



AA-Omniscience hallucination rate, 30 August 2026 snapshot. Lower is better. This metric rewards abstention, so a small model that declines often scores well without knowing much.

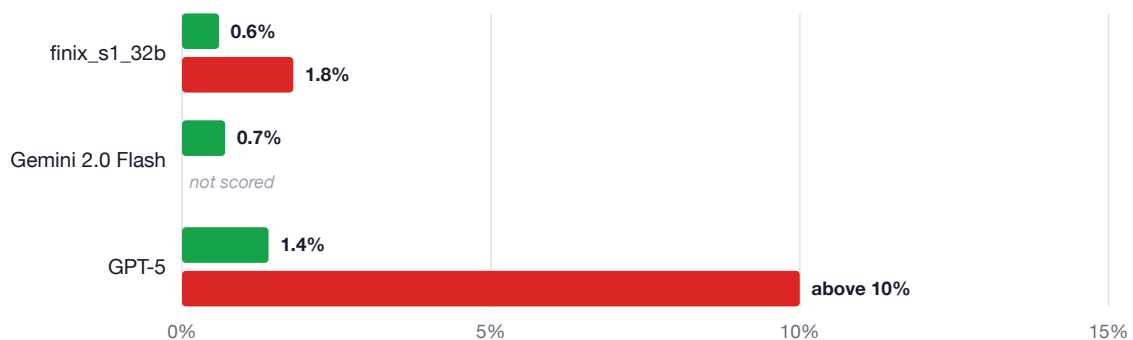
Which is why the rate alone is a bad way to pick a model. AA-Omniscience also publishes an **index** from –100 to 100 that combines knowledge with calibration — credit for being right, a penalty for guessing, credit for declining when uncertain. On that combined measure the frontier models do lead:



AA-Omniscience Index. 0 means a model is right as often as it is wrong. When the benchmark launched in November 2025, only 3 of 36 models scored above zero.

Why the old progress chart was wrong

The March version of this analysis showed a line falling from 21.8% to 0.7% and called it a 96% reduction. That line was real, but it was measuring the old Vectara dataset — a thousand short, simple articles. Rerun the same models on documents up to 32K tokens across law and medicine and the numbers move by an order of magnitude:



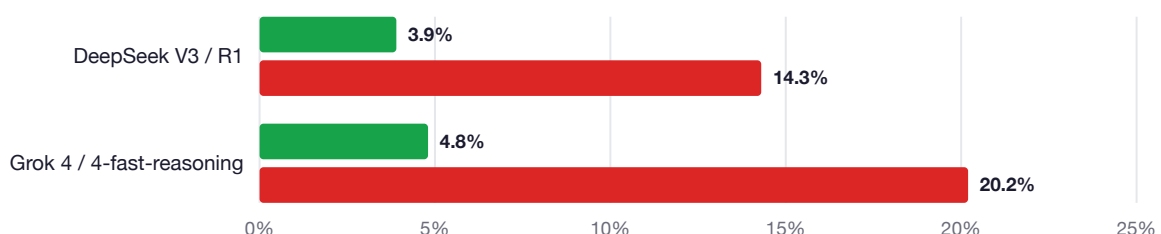
Old dataset (1,000 short articles) New dataset (7,700 articles, up to 32K tokens)

Vectara, both datasets. Gemini 2.0 Flash was not rescored on the new set. GPT-5's new-set figure is published as "above 10%" rather than an exact value; the bar is drawn at 10%.

The lesson is not that the progress was fake. It is that **a benchmark stops measuring progress as soon as models saturate it**, and by 2025 that benchmark was saturated. The sub-1% figures that got quoted everywhere described a test, not a capability.

The reasoning penalty

The most practical finding of 2026, and the least intuitive: turning on extended thinking makes a model **worse** at sticking to facts, even as it gets better at hard problems.



■ Base model ■ Reasoning variant

Vectara HHEM. Reasoning variants across the field run roughly 2–3× their base model's hallucination rate.

The plausible reading is that a longer chain of thought is more room to commit to a wrong premise and then build on it. Whatever the mechanism, it inverts the usual assumption: for a task where faithfulness matters more than cleverness, the reasoning mode is the wrong setting.

By contrast, **giving the model web access cuts hallucination by 73–86%** — far and away the largest single intervention anybody has measured. Retrieval beats reasoning, and it is not close.

Why does it actually happen?

The mechanism research has held up well, and this section is largely as it was. Causes sit at three levels: **data, training, and architecture**.

1. Data

Models are trained on internet-scale scrapes containing misinformation, outdated facts and internal contradictions. Where the training data holds two conflicting claims about the same thing, the model picks one on statistical grounds, not on truth. It also over-weights popular-but-wrong claims repeated millions of times, and under-weights obscure-but-correct facts that appear twice.

2. Training incentives — the root cause

OpenAI's September 2025 paper argues this is the *primary* driver: models are trained and scored in ways that **reward confident guessing over admitting uncertainty**. A student who always writes something down outscores one who leaves blanks, even when they are making it up.

Traditional benchmarks measure accuracy conditional on answering — not on the decision to answer.

The 2026 data is the strongest evidence yet that this is right. GPT-5.6 Sol guesses on 92.2% of what it does not know. That is not a model that failed to learn calibration; it is a model that learned exactly what the scoreboard rewards.

3. Architectural limits

The model does not "know" facts — it predicts the next likely token. Three mechanisms contribute:

- **Attention drift.** Attention concentrates on nearby tokens, so broader context thins out over a long output.
- **Near-miss probability.** A false statement that looks statistically like a true one gets assigned too much probability.
- **Inhibition-circuit failure.** Anthropic's interpretability work on Claude located circuits responsible for declining to answer. Hallucination happens when they are wrongly inhibited — the model recognizes a name, has no actual knowledge attached to it, and generates plausible detail instead of stopping.

4. Is it mathematically inevitable?

Two independent teams say yes. Xu et al. use computational learning theory to show that a model cannot learn all computable functions and will therefore hallucinate. Banerjee et al. reach the same conclusion from Gödel's First Incompleteness Theorem. Both find a non-zero hallucination probability at every stage of the pipeline, and a real trade-off between *consistency* (never producing invalid output) and *breadth* (producing diverse, useful output). Any model that generalizes past its training data either hallucinates or collapses into repetition.

What actually works

Ranked by measured effect, largest first:

1. **Give it sources.** Web access or retrieval cuts hallucination 73–86%. Nothing else comes close. The catch, from a 2025 Stanford study of legal RAG systems: a well-built pipeline can still fabricate citations, so span-level verification is still needed.
2. **Turn reasoning off for factual work.** Free, immediate, and worth 2–3× on grounded tasks.
3. **Fix the evaluations.** The OpenAI paper's own prescription, and the 2026 numbers make it urgent: while leaderboards reward confident guessing, models will guess. AA-Omniscience scoring abstention as a positive is the first widely-cited benchmark to actually do this.
4. **Uncertainty-aware training.** Penalize over- and under-confidence both. A 2025 NAACL study cut hallucination 90–96% on translation without hurting quality.
5. **Constrain the prompt.** A 2025 *Communications Medicine* study took GPT-4o from 53% to 23% in a clinical setting with prompt changes alone — explicit permission to say "I don't know" being the main one.
6. **Interpretability.** The longest-term route: find the circuits that decide whether to answer, and strengthen them.

Conclusion

Hallucination is not a glitch. It is the predictable output of how these systems are built, trained and scored. The model is not lying — it is predicting plausible text, and plausible text is sometimes false.

What is new in 2026 is that the comfortable story fell apart. Rates were never on a smooth path to zero; that impression came from a benchmark the models had outgrown. The current frontier knows more than any previous generation and is *less* willing to admit the edge of its knowledge. Meanwhile the proofs that some hallucination is structurally unavoidable have not been refuted.

There is real good news in here, and it is unglamorous: the largest improvement available to anyone is **attaching the model to sources and turning the clever reasoning off**. Both are free. Both are available today.

The takeaway for anyone using AI: verify anything that matters. A model that is right 65% of the time and guesses on almost two-thirds of the rest is a powerful tool. It is not a source of truth.

A note on these numbers

Benchmark figures here are drawn from Vectara's HHEM leaderboard and Artificial Analysis's AA-Omniscience, both of which update continuously; each chart states the snapshot date it reflects. Several figures reached me through aggregators reporting those leaderboards rather than from the leaderboards directly, and where the aggregators disagreed — Claude Fable 5's accuracy is reported as both 61% and 65% at different dates — I have used the most recent figure and said so. Treat single percentages as accurate to within a point or two, and the pattern as the real finding.

Sources

SOURCE	DATE
Artificial Analysis — <i>AA-Omniscience: Knowledge and Hallucination Benchmark</i> (arXiv 2511.13029) · artificialanalysis.ai/evaluations/omniscience	Nov 2025, data to Aug 2026
Vectara — <i>Introducing the Next Generation of Vectara's Hallucination Leaderboard</i>	19 Nov 2025
Vectara — HHEM leaderboard, new dataset	11 May 2026
BenchLM — AA-Omniscience hallucination-rate leaderboard	30 Aug 2026
Kalai, Nachum, Vempala, Zhang — <i>Why Language Models Hallucinate</i> (arXiv 2509.04664)	Sep 2025
Xu et al. — <i>Hallucination is Inevitable: An Innate Limitation of Large Language Models</i> (arXiv 2401.11817)	Jan 2024 / Feb 2025
Banerjee, Agarwal, Singla — <i>LLMs Will Always Hallucinate, and We Need to Live With This</i> (arXiv 2409.05746)	2025
Huang et al. — <i>A Survey on Hallucination in Large Language Models</i> (arXiv 2311.05232)	Nov 2023 / Nov 2024
Anthropic — interpretability research on Claude's internal representations	2025
<i>Communications Medicine</i> — adversarial hallucination in clinical decision support	2025
Stanford — study of legal RAG pipelines	2025
Columbia Journalism Review — citation accuracy across search models	Mar 2025
Stanford — AI Index	13 Apr 2026
AI-fabricated-citation case tracking (10 cases 2023 → 1,963 by Aug 2026)	Aug 2026

Viktor Brnclík · vbtronic.com · written with the help of AI tools Amp and Claude Code, and checked against the primary leaderboards.